program2

February 10, 2026

[1]:
```
'''Develop a program to Compute the correlation matrix to understand the␣
↪relationships between pairs of features.
Visualize the correlation matrix using a heatmap to know which variables have␣
↪strong positive/negative correlations.
Create a pair plot to visualize pairwise relationships between features. Use␣
↪California Housing dataset.'''
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
```

[2]:
```
# Step 1: Load the California Housing Dataset
california_data = fetch_california_housing(as_frame=True)
data = california_data.frame
print(data)
```

```
       MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0      8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1      8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2      7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3      5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4      3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85
…         …       …       …         …           …         …
20635  1.5603      25.0  5.045455   1.133333       845.0  2.560606     39.48
20636  2.5568      18.0  6.114035   1.315789       356.0  3.122807     39.49
20637  1.7000      17.0  5.205543   1.120092      1007.0  2.325635     39.43
20638  1.8672      18.0  5.329513   1.171920       741.0  2.123209     39.43
20639  2.3886      16.0  5.254717   1.162264      1387.0  2.616981     39.37

       Longitude  MedHouseVal
0        -122.23        4.526
1        -122.22        3.585
2        -122.24        3.521
3        -122.25        3.413
4        -122.25        3.422
…          …           …
20635    -121.09        0.781
20636    -121.21        0.771
```

```
20637    -121.22        0.923
20638    -121.32        0.847
20639    -121.24        0.894

[20640 rows x 9 columns]
```

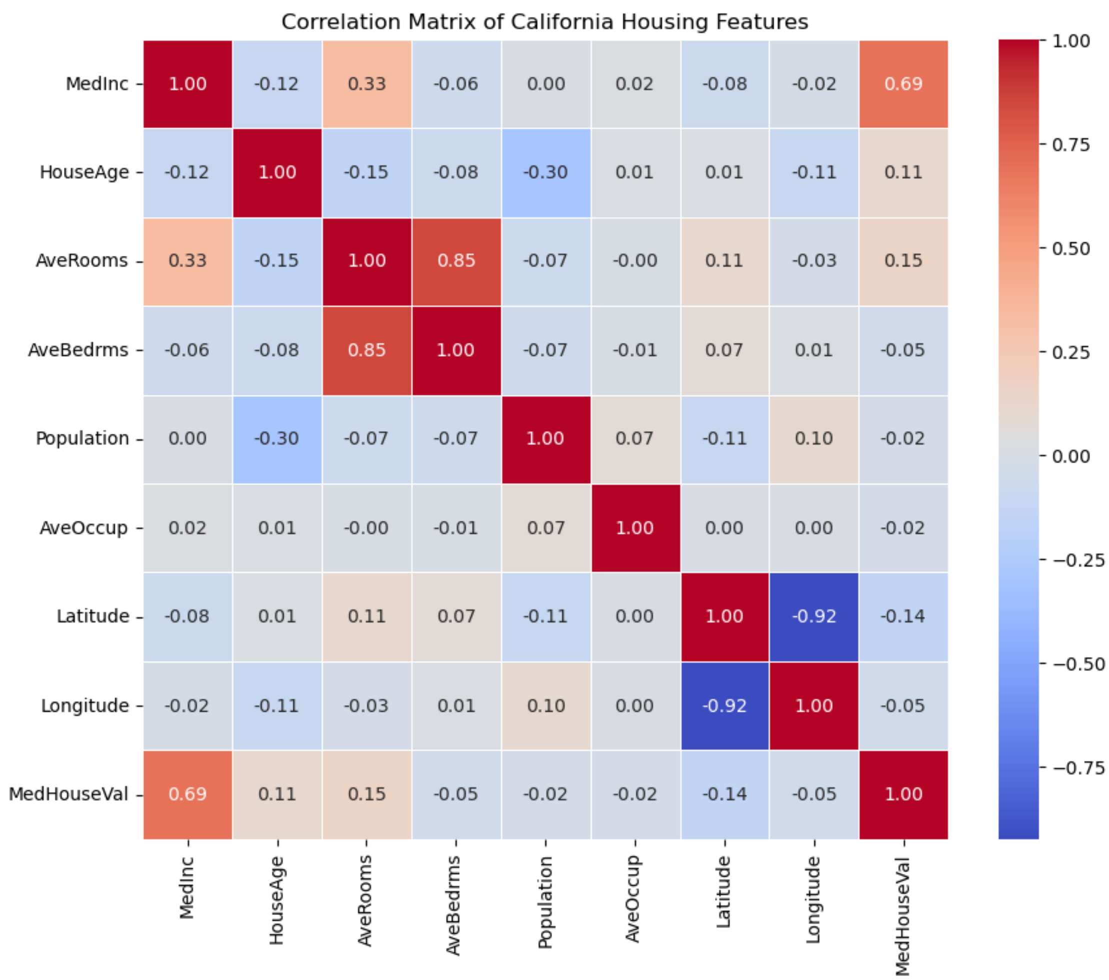
[3]:

```python
# Step 2: Compute the correlation matrix
correlation_matrix = data.corr()
print(correlation_matrix)
```

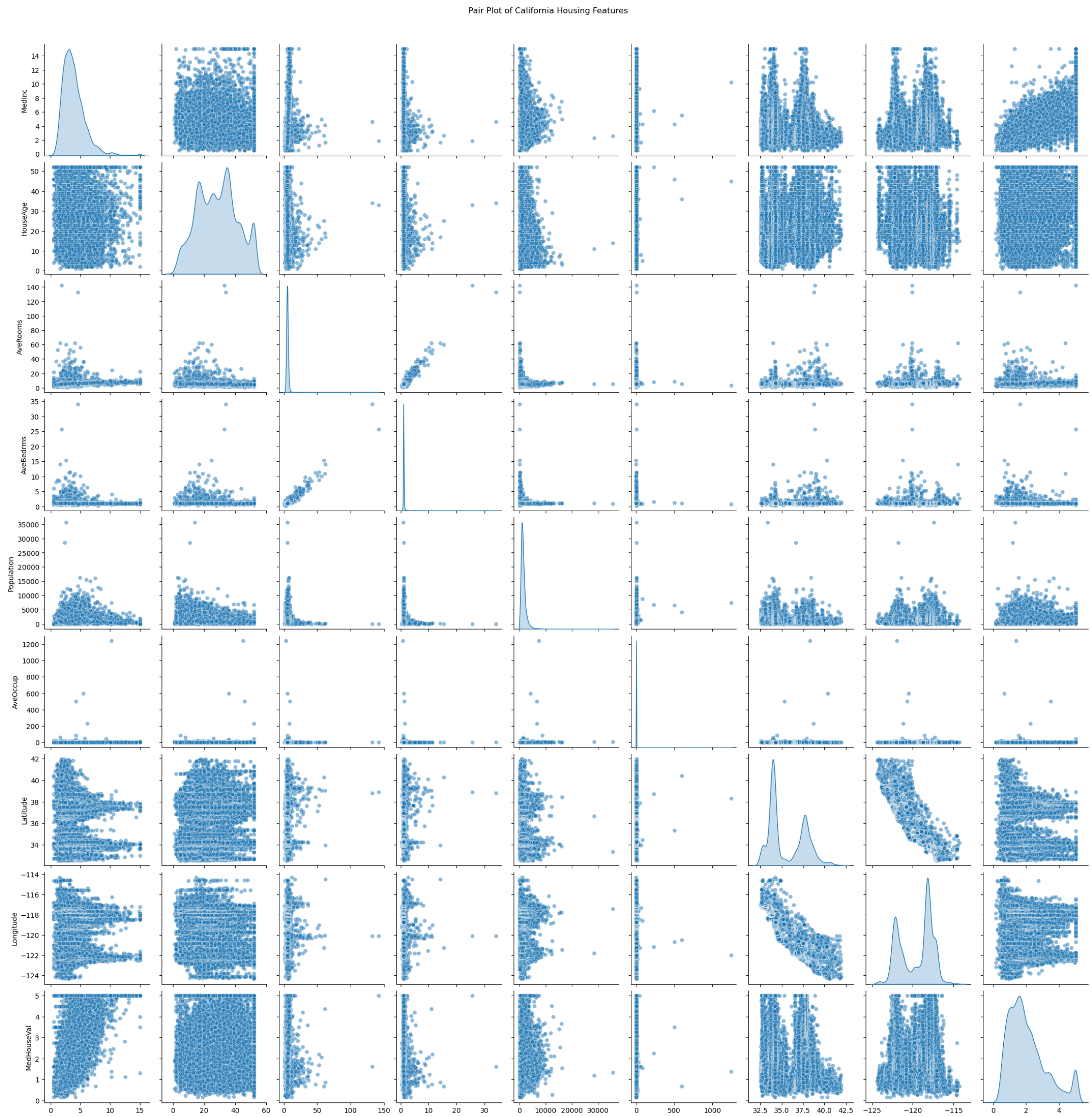
```
               MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  \
MedInc       1.000000 -0.119034  0.326895  -0.062040    0.004834  0.018766
HouseAge    -0.119034  1.000000 -0.153277  -0.077747   -0.296244  0.013191
AveRooms     0.326895 -0.153277  1.000000   0.847621   -0.072213 -0.004852
AveBedrms   -0.062040 -0.077747  0.847621   1.000000   -0.066197 -0.006181
Population   0.004834 -0.296244 -0.072213  -0.066197    1.000000  0.069863
AveOccup     0.018766  0.013191 -0.004852  -0.006181    0.069863  1.000000
Latitude    -0.079809  0.011173  0.106389   0.069721   -0.108785  0.002366
Longitude   -0.015176 -0.108197 -0.027540   0.013344    0.099773  0.002476
MedHouseVal  0.688075  0.105623  0.151948  -0.046701   -0.024650 -0.023737

             Latitude  Longitude  MedHouseVal
MedInc      -0.079809  -0.015176     0.688075
HouseAge     0.011173  -0.108197     0.105623
AveRooms     0.106389  -0.027540     0.151948
AveBedrms    0.069721   0.013344    -0.046701
Population  -0.108785   0.099773    -0.024650
AveOccup     0.002366   0.002476    -0.023737
Latitude     1.000000  -0.924664    -0.144160
Longitude   -0.924664   1.000000    -0.045967
MedHouseVal -0.144160  -0.045967     1.000000
```

[4]:

```python
# Step 3: Visualize the correlation matrix using a heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f',␣
 ↪linewidths=0.5)
plt.title('Correlation Matrix of California Housing Features')
plt.show()
```



[5]:
```
# Step 4: Create a pair plot to visualize pairwise relationships
sns.pairplot(data, diag_kind='kde', plot_kws={'alpha': 0.5})
plt.suptitle('Pair Plot of California Housing Features', y=1.02)
plt.show()
```


Pair Plot of California Housing Features
MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
Latitude
Longitude
MedHouseVal

[ ]: